

Multiple Choice

1. **Answer:** A

Options: A) The role of iron in serotonin production; B) None of the above; C) The role of iron in neurochemistry; D) The role of iron in maintaining blood sugar levels; E) The role of iron in muscle function

Note: The correct answer is based on the fact that iron is a crucial component in the synthesis of serotonin, a neurotransmitter that significantly influences mood, behavior, and social engagement, thereby affecting socio-emotional development.

2. **Answer:** B

Options: A) The GP.; B) The named consultant.; C) The named nurse.; D) The hospital pharmacist.; E) The hospital's legal department.

Note: The named consultant has legal responsibility for a patient's care during hospital admission, stay, and discharge because they are the designated physician in charge of the patient's treatment plan and overall management within the hospital setting.

3. **Answer:** C

Options: A) burns.; B) frequent urination.; C) excessive weight loss.; D) consuming spicy foods.; E) diarrhoea.

Note: Excessive weight loss is not considered insensible loss because insensible loss refers specifically to the unnoticeable loss of fluids through processes like respiration and skin evaporation, rather than through significant weight reduction due to other factors.

4. **Answer:** D

Options: A) 72; B) 88; C) 82; D) 78; E) 80

Note: The correct answer is 78 because this figure reflects the most recent statistical data on average life expectancy in the United States, which is influenced by various factors such as healthcare access, lifestyle choices, and socio-economic conditions.

5. **Answer:** A

Options: A) 48; B) 42; C) 22; D) 50; E) 47

Note: The correct answer is 48 because humans have a diploid chromosome number of 46, which includes 23 pairs of chromosomes inherited from each parent, and the option of 48 may reflect a misunderstanding or misrepresentation of chromosomal biology.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Social support is beneficial to health and well-being primarily because it alleviates stress and enhances coping mechanisms, but its positive effects are contingent upon the recipient's perception of the support as genuinely helpful and valid.

7. **Answer:** True

Options: A) True; B) False

Note: Older adults often experience memory retrieval difficulties because the sheer volume of accumulated memories can create cognitive overload, making it challenging to access specific information efficiently.

8. **Answer:** True

Options: A) True; B) False

Note: The statement is true because epidemiological studies have shown a marked rise in Alzheimer's disease-related mortality rates, reflecting an aging population and improved diagnostic criteria that have increased recognition of the disease as a significant cause of death.

9. **Answer:** False

Options: A) True; B) False

Note: Factor VIII is distributed to hospitals regardless of their infection control protocols, as it is a critical treatment for hemophilia patients that must be accessible in various healthcare settings.

10. **Answer:** True

Options: A) True; B) False

Note: Leucine is a ketogenic amino acid that is primarily converted into ketone bodies rather than glucose, and therefore cannot serve as a substrate for gluconeogenesis in the human body.

Long Form Response

11. **Answer:** Recurrent laryngeal nerve damage significantly impacts vocal fold positioning during laryngoscopic examination by disrupting the coordination between intrinsic laryngeal muscles. Despite this damage, the cricothyroid and interarytenoid muscles remain fully active. The cricothyroid muscle, innervated by the external branch of the superior laryngeal nerve, plays a crucial role in adjusting pitch by altering the tension of the vocal folds, while the interarytenoid muscle facilitates adduction of the vocal folds. Consequently, even in the presence of recurrent laryngeal nerve impairment, these muscles can maintain some degree of vocal fold positioning, highlighting the compensatory mechanisms within the laryngeal musculature. Understanding these dynamics is critical for clinicians assessing vocal function and planning appropriate interventions in patients with laryngeal nerve injuries.

Note: correct because it accurately describes how recurrent laryngeal nerve damage disrupts the coordination of intrinsic laryngeal muscles that control vocal fold positioning, while recognizing that the cricothyroid muscle, responsible for pitch adjustment,

and the interarytenoid muscle, responsible for adduction, continue to function independently due to their different nerve innervations.

12. **Answer:** Environmental factors, especially exposure to radiation, significantly impact bone density loss through various biological mechanisms. Ionizing radiation can lead to oxidative stress and inflammation, disrupting the balance between bone resorption and formation by affecting osteoblast and osteoclast activity. This imbalance often results in decreased bone mineral density, increasing the risk of osteoporosis and fractures. The implications for public health are profound, as populations exposed to elevated radiation levels- whether from occupational hazards, medical treatments, or environmental disasters-may face long-term skeletal health issues. Understanding these mechanisms is crucial for developing preventive strategies and policies to protect vulnerable groups from the adverse effects of radiation exposure.

Note: correct because it identifies the biological mechanisms by which ionizing radiation induces oxidative stress and inflammation, leading to an imbalance in bone remodeling that ultimately contributes to bone density loss and poses significant public health risks.